

Hitec University Taxila, Department of Computer Engineering

Lab Manual for Mobile Application Development

Lab-12 Android Networking using Volley Library

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1 Introduction

Volley is an HTTP library that makes networking for Android apps easier and most importantly, faster. Volley offers the following benefits:

- Automatic scheduling of network requests
- Multiple concurrent network connections
- Transparent disk and memory response caching with standard HTTP cache coherence
- Cancellation request API. You can cancel a single request, or you can set blocks or scopes of requests to cancel.
- Strong ordering that makes it easy to correctly populate your UI with data fetched asynchronously from the network.

2 Activity Time boxing

Table 1: Activity Time Boxing

Task#	Activity Name	Activity time	Total Time
1	Introduction	60mins	60min
2	Walk Through Task	60mins	60min
3	Evaluation Task	60min	60min

3 Objectives of the experiment

- To get basic understanding regarding android volley networking
- To get basic understanding regarding how to use volley networking in HTTP connections
- To get to know about the benefits related to android volley networking.

4 Walk through Task

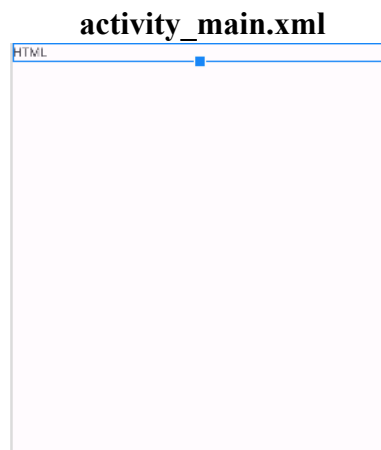
Creating New Activity

Step 1: Create Activities:

Create a new project just like you have created in your previous labs and create one more empty activity.

Step 2: Design XMLs of Activity:

Designs of activity are given below, just use the given xml code which is given in this step.



```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/responce_tv"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="HTML" />
</LinearLayout>

```

Step 3: Add Volley Library in gradle file

Under gradle>build.gradle

Add this and press sync gradle

```

dependencies {
    .
    .
    implementation("com.android.volley:volley:1.2.0")
    .
    .
}

```

Step 4: Write “MainVolleyNetwork.java” class

```

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.util.Log;
import android.widget.TextView;

import com.android.volley.Request;
import com.android.volley.RequestQueue;
import com.android.volley.Response;
import com.android.volley.VolleyError;
import com.android.volley.toolbox.StringRequest;
import com.android.volley.toolbox.Volley;

public class MainVolleyNetwork extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main_volley_network);

        TextView tv = findViewById(R.id.responce_tv);

        // Instantiate the RequestQueue.
        RequestQueue queue = Volley.newRequestQueue(this);
        String url ="https://www.york.ac.uk/teaching/cws/wws/webpage1.html";
    }
}

```

```

// Request a string response from the provided URL.
StringRequest stringRequest = new StringRequest(Request.Method.GET, url,
    new Response.Listener<String>() {
        @Override
        public void onResponse(String response) {
            Log.d("Main", "Response is" + response);
            tv.setText(response);
        }
    },
    new Response.ErrorListener() {
        @Override
        public void onErrorResponse(VolleyError error) {
            Log.d("Main", "No response");
        }
    }
);

// Add the request to the RequestQueue.
queue.add(stringRequest);
}
}

```

Step 3: Initialize permission in AndroidManifest.xml

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.INTERNET" />

    <application
    ...

```

Step 6: Now Run Project

Click “Run” or press “Shift + F10”.

